

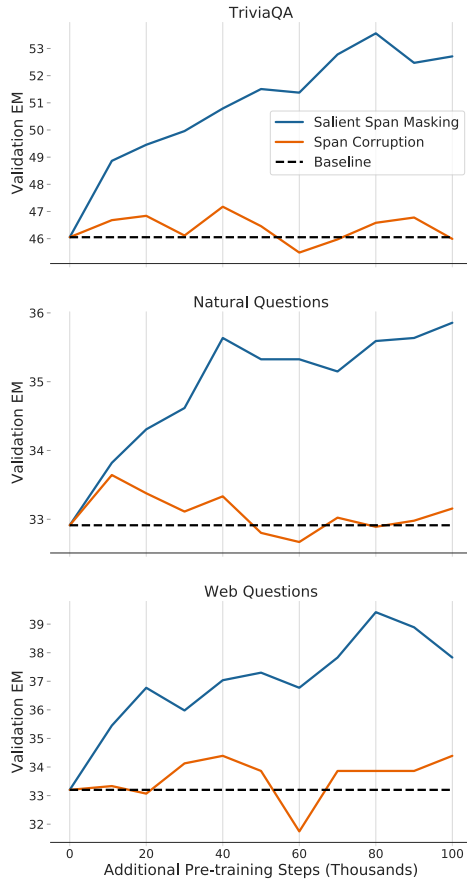


Figure 2: Comparing additional pre-training using either salient span masking (SSM) or span corruption (SC). We further pre-trained T5.1.1-XXL on the Wikipedia sentence dataset from Guu et al. (2020) with each objective, fine-tuning on a mixture of our three closed-book QA tasks every 10,000 steps. For each fine-tuning run, we report the maximum exact match score achieved on the validation set over 10,000 steps of fine-tuning.

gains achieved were attributable to the use of a more task-specific dataset (pre-split into sentences that are known to contain at least one entity) or if the SSM objective itself was critical. As illustrated in fig. 2, the SSM objective is clearly an important ingredient in the improved performance; we saw no significant improvement versus the baseline T5 model when using SC.

Fine-Tuning On All Question Answering Tasks

The text-to-text framework used by T5 makes it simple to train multitask models simply by supplying a different task-specific prefix for each task and concatenating all of the constituent datasets. Since all of the question answering tasks we consider in this study follow the same basic structure, we were hopeful that training on a multitask

mixture of Natural Questions, WebQuestions, and TriviaQA would improve performance due to the additional supervised data. While multitask training improved performance on the Natural Questions by 0.5, it produced slightly worse results on the other tasks.

Randomly Sampling Answers For Natural Questions

In the open-domain variant of Natural Questions, the model is only trained to generate a single answer at a time. For the results presented in the main text, when a question was annotated with multiple answers, we simply trained the model on the first annotated answer. We also experimented with sampling a random answer from the set of possible answers for pre-training and found that it did not affect performance.